Characterizing Grokking via Topological Data Analysis in a Small Transformer Model

Gabriel Ribeiro[1], Petrônio Silva[2], Pedro H. Barros[1], and Heitor S. Ramos[1]

[1] Universidade Federal de Minas Gerais (UFMG), Belo Horizonte, Brazil
[2] Instituto Federal do Norte Minas Gerais, Januária, Minas Gerais, Brazil
gabriel.ribeiro@dcc.ufmg.br

Abstract. The grokking phenomenon, in which neural networks transition from memorization to sudden generalization after prolonged training, remains an important area of ongoing research in deep learning. While prior work has focused on mechanistic or symmetry-based explanations, less attention has been given to the global structure of learned representations. In this work, we propose a novel perspective by leveraging Topological Data Analysis to investigate the geometric and topological evolution of representations in a small transformer model trained on modular arithmetic tasks. Our approach combines persistent homology and intrinsic dimension estimation to track how representation spaces evolve across different layers during training. We show that successful generalization frequently coincides with a structural simplification of the representation space, characterized by a reduction in intrinsic dimension and a stabilization of topological features. In contrast, models that fail to generalize exhibit higher-dimensional and unstable topological structures. These findings suggest that grokking can be understood as a topological shift toward simpler and more organized manifolds, offering a new algebraic-topological interpretation of generalization in deep neural networks.

Keywords: Grokking · Topological Data Analysis · Homology

1 Introduction

A fundamental objective across all domains of machine learning is achieving high predictive accuracy on both training and unseen validation data. While the optimization dynamics of deep neural networks are inherently stochastic, severe overfitting is universally regarded as a condition to be avoided. Consequently, the complex relationship between memorization and generalization remains one of the most significant open questions in the field. Demystifying these internal dynamics, and discovering whether generalization can be systematically induced, is crucial for advancing our theoretical understanding of deep learning.

1.1 Contextualization

In 2022, OpenAI published a paper on generalization beyond overfitting [8] in a model trained on modular operations, coining the term "grokking", derived from the verb *to grok*, meaning to understand something profoundly. This phenomenon describes a scenario where a model heavily overfits the training data for many epochs, but eventually overcomes this memorization phase to achieve perfect generalization on the validation set.

Because of its counterintuitive nature, significant efforts have been made to understand the mechanics behind this phenomenon. Prior work has primarily investigated how these models generalize modular operations through the lens of mechanistic interpretability [7], or by analyzing intrinsic task symmetries [2]. However, purely computational or circuit-based approaches often miss the broader structural and geometric reorganization occurring within the model's representation space.

In this work, we analyze the grokking phenomenon through the lens of Topological Data Analysis (TDA), seeking to identify topological and geometrical indicators that arise during the grokking process. By examining the evolving shape of the data representations, we aim to provide a rigorous topological characterization of the grokking effect.

1.2 Contributions

Specifically, our main contributions are as follows:

- **Layer-Wise Topological Analysis:** We provide a granular analysis of how the geometric shape of data representations evolves across distinct components of a transformer model (embedding, decoder blocks, and linear layers) during the transition from memorization to generalization.
- **Topological Characterization of Grokking:** We identify key structural shifts and topological markers (such as the emergence or collapse of specific persistent homology classes) that correspond to the onset of the grokking phenomenon, offering a new geometric perspective on delayed generalization.

1.3 Paper Outline

The remainder of this paper is structured as follows. Section 2 discusses related work in the fields of mechanistic interpretability and representation geometry. Section 3 details the background of TDA. Section 4 outlines our methodology, including the architecture, training regime, and mathematical pipeline used to extract topological invariants. Section 5 presents our empirical results and the observed topological markers of grokking, and Section 6 concludes with our final remarks and directions for future research.

2 Methodology

Our analysis proceeds through a six-stage pipeline: task and dataset definition, model training, representation extraction, intrinsic dimension estimation, simplicial complex construction, and persistent homology computation.

2.1 Task Formulation and Datasets

Task formulation. We study two modular arithmetic tasks over $\mathbb{Z}_p$ with prime modulus $p = 97$, formulated as next-token prediction following Power et al. [8]. Each input is the four-token sequence $\langle\, a \circ b \;= \rangle$, where $a, b \in \{0, \ldots, p-1\}$ and $\circ \in \{+, \times\}$. The model's vocabulary has size $|\mathcal{V}| = p + 2 = 99$, comprising the p residues, one operator token, and one equality token. The training objective is to predict, at the position immediately following the equality token, the single token corresponding to

$$y = (a \;\circ\; b) \bmod p \;\in\; \{0, \ldots, 96\}. \tag{1}$$

The loss is the cross-entropy between the model's output distribution over $\mathcal{V}$ and the one-hot encoding of y; accuracy is the fraction of examples for which the model's argmax equals y.

Modular addition. The full dataset consists of all $p^2 = 9{,}409$ ordered pairs (a, b). For example, the input $\langle\, 45 \;+\; 72 \;= \rangle$ should elicit $y = (45 + 72) \bmod 97 = 20$. The task defines an abelian group on $\mathbb{Z}_{97}$, admitting an exact decomposition via discrete Fourier analysis over the group [7], which the model must discover implicitly through gradient descent.

Modular multiplication. The dataset again contains all $p^2 = 9{,}409$ ordered pairs. For example, $\langle\, 15 \;\times\; 9 \;= \rangle$ should elicit $y = (15 \times 9) \bmod 97 = 38$. The non-zero residues $\mathbb{Z}_{97}^*$ form a cyclic group of order $p-1 = 96$ under multiplication; the case $a = 0$ or $b = 0$ trivially yields $y = 0$. Multiplication is empirically harder to grok than addition, requiring more training epochs before generalization occurs [8].

Train/validation splits. For each task and run, the p^2 examples are randomly partitioned into a training set of fraction $q \in \{10\%, 15\%, 20\%, 25\%, 30\%\}$ and a held-out validation set of the remaining $1 - q$ fraction [3]. Varying q controls the difficulty of generalization: smaller fractions provide less coverage of the input space, making it harder for the model to discover the underlying algebraic structure.

2.2 Model Architecture and Training

We use the decoder-only transformer of Power et al. [8] without modification: two transformer blocks, each with 4-head self-attention and a feedforward sublayer of hidden dimension 512, preceded by a 128-dimensional token-plus-positional embedding layer and followed by a linear projection to $\mathbb{R}^{|\mathcal{V}|}$. Training uses AdamW [5] with learning rate 10^{-3}, and weight decay $\lambda = 1.0$, with full-batch gradient descent for $10^6 + 1$ epochs. Validation accuracy is evaluated at each epoch without gradient updates.

[3] No data augmentation or curriculum ordering is applied.

2.3 Representation Extraction

At regular checkpoint epochs t, we perform a forward pass over the *entire* dataset with gradients disabled and record activations at the position of the equality token from four components: (1) the token-plus-positional embedding layer, $A_t^{(0)} \in \mathbb{R}^{n\times128}$; (2) the first decoder block output, $A_t^{(1)} \in \mathbb{R}^{n\times128}$; (3) the second decoder block output, $A_t^{(2)} \in \mathbb{R}^{n\times128}$; and (4) the pre-softmax linear projection, $A_t^{(3)} \in \mathbb{R}^{n\times99}$, where $n = p^2 = 9{,}409$. Each matrix $A_t^{(\ell)}$ is treated as a point cloud in its ambient space, with one point per input example. The equality-token position is used throughout because it is the position from which the output distribution is computed, and is therefore the most direct representation of the model's belief about the answer at checkpoint t.

2.4 TDA Pipeline

Intrinsic dimension estimation. For each layer ℓ and checkpoint t, we estimate the intrinsic dimension $\hat{d}_t^{(\ell)}$ of $A_t^{(\ell)}$ using the MLE of Levina and Bickel [3] (Equation 4, $k = 10$ nearest neighbors). Note that, this estimate serves two objectives: as a standalone geometric descriptor tracked over training, and as the target dimension for the subsequent reduction step.

Dimensionality reduction. We apply UMAP [6] to reduce $A_t^{(\ell)}$ to $\tilde{A}_t^{(\ell)} \in \mathbb{R}^{n\times2\hat{d}}$, with target dimension $2\hat{d}$ motivated by the Strong Whitney Embedding Theorem (Section 3.2). As discussed in Section 3.2, UMAP is a heuristic that preserves local connectivity but provides no formal topological guarantees; the reported Betti numbers are exact invariants of the constructed complex but should be interpreted as indicative of structural trends with respect to the underlying activation manifold.

Simplicial complex construction. On the reduced $\tilde{A}_t^{(\ell)}$, we construct a graph using the algorithm of Liu et al. [4]: an edge (x_i, x_j) is included if $\|x_i - x_j\|_2 \le \varepsilon$, where ε is the 95th percentile of distances to the $k = 31$-th nearest neighbor across all points. The graph is expanded into a full simplicial complex via GUDHI's simplex tree [9], adding 2- and 3-simplices wherever the corresponding cliques exist. The choices $k = 31$ and the 95th-percentile threshold are heuristic [4].

Persistent homology. We compute persistent homology via GUDHI's boundary matrix reduction [9] for dimensions $k \in \{0, 1, 2\}$. From each persistence diagram $\mathcal{D}_t^{(\ell,k)}$ we extract $\beta_t^{(\ell,k)}$ (see Section 3.1) as the count of pairs with persistence exceeding 5% of the maximum observed distance (noise threshold $\tau = 0.05$). We restrict to $k \le 2$ because $\mathbb{Z}_{97} \cong$ the 97th roots of unity embeds naturally in a 2-torus $\mathbb{T}^2$, whose homology is trivial (i.e., equal to zero) above dimension 2; higher-dimensional features are negligible in our experiments.

[4] A sensitivity analysis over these choices is needed to establish the robustness of the reported signatures, and it remains future work

2.5 Evaluation Protocol

We compare the longitudinal topological time series $\{\beta_t^{(\ell,k)}, \hat{d}_t^{(\ell)}\}$ across all checkpoints t, layers $\ell \in \{0,1,2,3\}$, and dimensions $k \in \{0,1,2\}$, for each task and training fraction q. We evaluate three hypotheses:

- **(H1) Betti stabilization.** In generalizing conditions, $\beta_t^{(\ell,0)}$ converges to a stable value as validation accuracy increases.
- **(H2) Layer-specific directionality.** Upon generalization, $\beta_t^{(\ell,0)}$ decreases in the decoder layers and increases in the linear layer.
- **(H3) Intrinsic dimension reduction.** Generalizing conditions exhibit a lower converged $\hat{d}_t^{(\ell)}$ across all layers than non-generalizing conditions.

3 Background on Topological Data Analysis

TDA extracts structural, shape-based features from data that are invariant to continuous deformations and robust to noise [1]. Figure 1 summarizes the pipeline used in this work: a metric point cloud is covered by a filtered simplicial complex, from which persistent homology is computed and summarized as a set of Betti numbers. We formalize each component below.

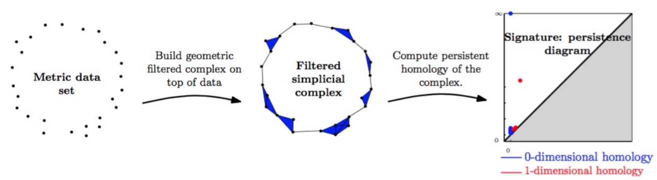

Fig. 1: Illustration of a typical workflow in Topological Data Analysis

3.1 Simplicial Complexes, Filtrations, and Homology

Simplicial complexes. A *k-simplex* is the convex hull of $k+1$ affinely independent points: a 0-simplex is a vertex, a 1-simplex is an edge, a 2-simplex is a triangle, and a 3-simplex is a tetrahedron. A *simplicial complex* K is a finite collection of simplices closed under the face relation: if $\sigma \in K$ and τ is a face of σ, then $\tau \in K$. Figures 2 and 3 illustrate two canonical examples: a three-dimensional complex (a solid tetrahedron) and a two-dimensional triangulation of a toroidal surface. A *triangulation* decomposes a surface into non-overlapping triangles such that adjacent triangles share exactly one edge or vertex. The torus is directly relevant to our setting: the cyclic group $\mathbb{Z}_p$ for $p \in \mathbb{N}$ underlying our arithmetic tasks is embeddable in a 2-torus $\mathbb{T}^2 = S^1 \times S^1$, which constrains the expected homological structure of the learned representations.

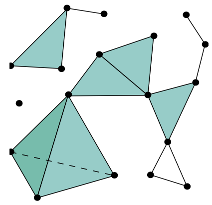

Fig. 2: Three dimensional Simplicial Complex, because its higher simplex has dimension 3, which is the tetrahedron.

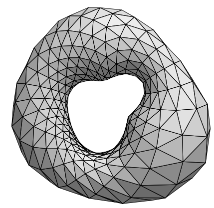

Fig. 3: A triangulation can be understood as a two dimensional Simplicial Complex, since its higher dimensional simplex are triangles.

Filtrations. Given a set $X = \{x_1, \dots, x_n\} \subset \mathbb{R}^D$, a *filtration* constructs a nested sequence of simplicial complexes

$$\emptyset = K_0 \subseteq K_1 \subseteq \cdots \subseteq K_m = K \tag{2}$$

indexed by a non-decreasing scale parameter $\varepsilon_0 \leq \cdots \leq \varepsilon_m$. In the *Vietoris–Rips* construction, $\{x_{i_0}, \dots, x_{i_k}\}$ spans a k-simplex at scale ε if and only if all pairwise distances are at most ε. As ε grows, isolated points are connected into edges, triangles, and higher simplices, as depicted in Figure 1 (center). For small ε the complex is disconnected; for large ε all topology collapses into a single contractible component. In our pipeline we use an adaptive k-nearest-neighbor variant, where an edge (x_i, x_j) is included if their distance falls within the 95th percentile of distances to the k-th nearest neighbor across X.

Homology and Betti numbers. The *boundary operator* $\partial_k : C_k \to C_{k-1}$ maps each k-simplex to the formal sum of its $(k-1)$-faces, satisfying $\partial_{k-1} \circ \partial_k = 0$. The *k-th homology group*

$$H_k(K; \mathbb{Z}_2) \;=\; \ker(\partial_k) \,/\, \operatorname{im}(\partial_{k+1}) \tag{3}$$

isolates k-dimensional cycles that do not bound any $(k+1)$-dimensional region. The *k-th Betti number* $\beta_k = \operatorname{rank}(H_k)$ counts these independent holes: β_0 counts connected components, β_1 counts independent loops, and β_2 counts enclosed voids. For example, the tetrahedron and torus in Figure 2 and 3, the Betti vectors are $(1, 0, 1)$ and $(1, 2, 1)$ respectively, confirming that these invariants capture the qualitatively different topological structure of the two shapes.

3.2 Intrinsic Dimension Estimation

A topological space M is a *d-dimensional manifold* if every point $x \in M$ has a neighborhood homeomorphic to an open subset of $\mathbb{R}^d$. The *intrinsic dimension* (ID) of a dataset $X \subset \mathbb{R}^D$ is the smallest d such that X is concentrated on or near a d-dimensional smooth submanifold of $\mathbb{R}^D$ — the *manifold hypothesis*. In practice, this is operationalized through local volume scaling: if the data locally resembles a d-dimensional manifold, the number of neighbors within a ball of radius r grows proportionally to r^d. We estimate ID using the Maximum Likelihood Estimator of Levina and Bickel [3], which formalizes this scaling behavior. Given distances $r_1(x) \leq \cdots \leq r_k(x)$ from x to its k nearest neighbors,

$$\hat{d}_k(x) = \left[\frac{1}{k-1} \sum_{j=1}^{k-1} \log \frac{r_k(x)}{r_j(x)} \right]^{-1}, \tag{4}$$

with the global estimate obtained by averaging over all $x \in X$.

We use $\hat{d}_k$ for two purposes in our pipeline. First, as a standalone geometric descriptor tracked over training (Section 4.2). Second, as the target dimension for dimensionality reduction prior to persistent homology computation: activation matrices are reduced to $\mathbb{R}^{2\hat{d}}$ via UMAP [6], with the target dimension motivated by the Strong Whitney Embedding Theorem [10].

Theorem 1 (Strong Whitney's Embedding Theorem [10]). *Any smooth real m-dimensional manifold (required also to be Hausdorff and second-countable) can be smoothly embedded in the real $2m$-space, $\mathbb{R}^{2m}$, if $m > 0$.*

Whitney's theorem guarantees the *existence* of a topology-preserving embedding into $\mathbb{R}^{2d}$; it does not guarantee that UMAP finds it. UMAP is employed here as a heuristic, chosen for its strength in preserving local connectivity structure [6], the property most relevant to the simplicial complexes constructed downstream. The Betti numbers computed from the resulting complex are exact invariants *of that complex*; the approximation lies in whether the complex faithfully represents the topology of the underlying activation manifold. The consistency of the observed topological trends across all ten experimental conditions (two tasks, five training fractions) provides empirical evidence that the pipeline captures genuine structural changes rather than reduction artifacts. Nevertheless, the reported Betti numbers should be interpreted as indicative of structural trends rather than exact topological invariants of the activation manifold.

4 Results

4.1 Betti Number Evolution

Modular addition — generalizing conditions. Figures 4 and 5 show the evolution of β_0, β_1, and β_2 for $q \in \{30\%, 20\%\}$. A consistent pattern accompanies the surge in validation accuracy: in the first decoder block, β_0 decreases monotonically, indicating association of disjoint activation clusters into fewer coherent connected components; in the linear projection layer, β_0 increases and stabilizes, indicating the formation of well-separated output clusters corresponding to the $p = 97$ target classes. Higher-dimensional Betti numbers β_1 and β_2 remain near zero throughout, confirming that the relevant topological dynamics are captured by β_0 alone.

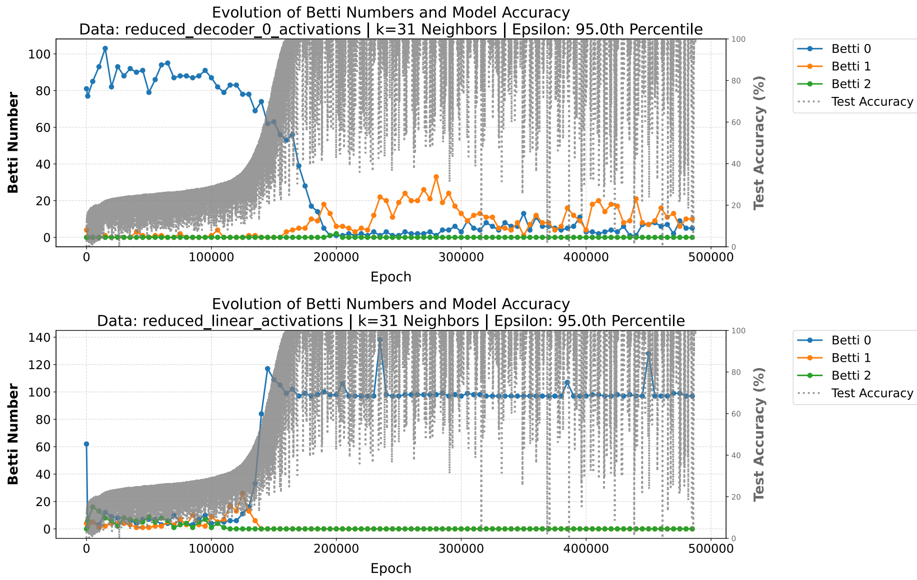

Fig. 4: **30% of training data.** Evolution of the Betti numbers for the first decoder and linear blocks. Notice that as validation accuracy rises, the number of connected components decreases for the first decoder block, while the opposite occurs in the linear block.

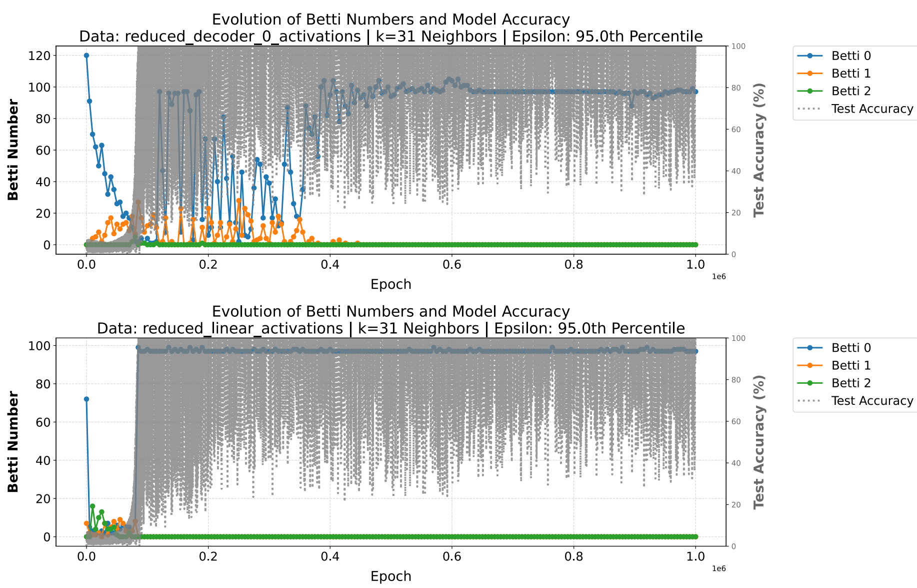

Fig. 5: **20% of training data.** The same behavior described in Fig. 4 can be observed here.

Modular addition — non-generalizing conditions. Figure 6 shows the corresponding evolution for $q \in \{15\%\}$. In the first decoder block, β_0 collapses more steeply and irregularly without reaching any stable value; in the linear layer, β_0 exhibits persistent high-amplitude oscillations throughout training, never converging. Overall, the 15% condition does not exhibit convergence behavior in either representation, in contrast to the stabilizing dynamics observed under generalizing conditions. This unstable behavior is the most visually distinctive signature of failed generalization and is consistent across both non-generalizing fractions, supporting **H1** and **H2**.

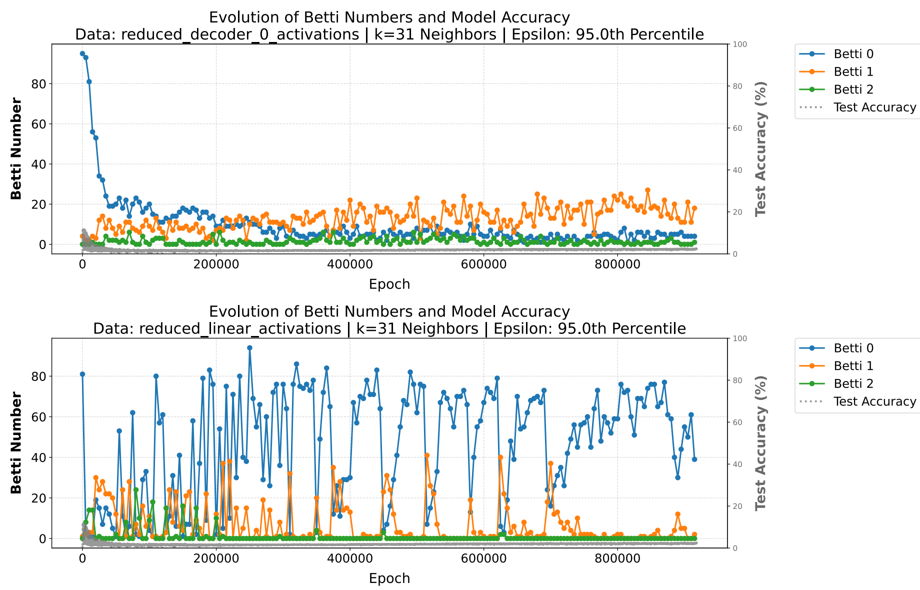

Fig. 6: **15% of training data.** Failure to generalize resulted in spikes in connected components within the linear layer and a steeper descent of connected components in the first decoder layer.

Modular multiplication — generalizing conditions. Figure 7 shows results for $q = 30\%$ on the multiplication task. The qualitative pattern is similar to the addition task, decoder β_0 decreases, linear β_0 converges ascending, despite generalization occurring substantially faster, confirming that the topological signatures are associated with the onset of generalization instead any particular training epoch. Figure 8 shows that at $q = 20\%$ the same convergence is observed in the linear layer, and the decoder converges to a higher number of connected components than in the addition task.

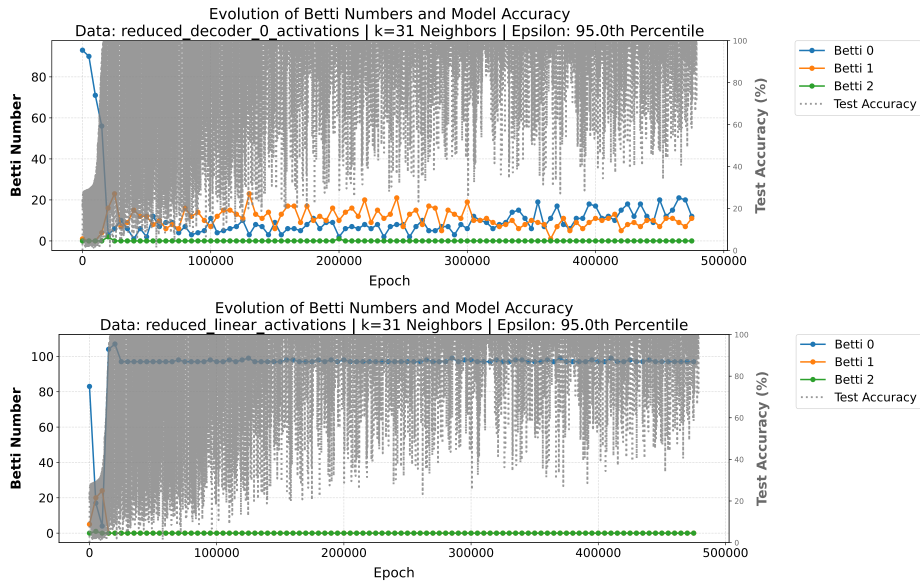

Fig. 7: **30% of training data.** Although generalization occurred much faster, the Betti numbers converged in the linear layer and decreased in the first decoder layer.

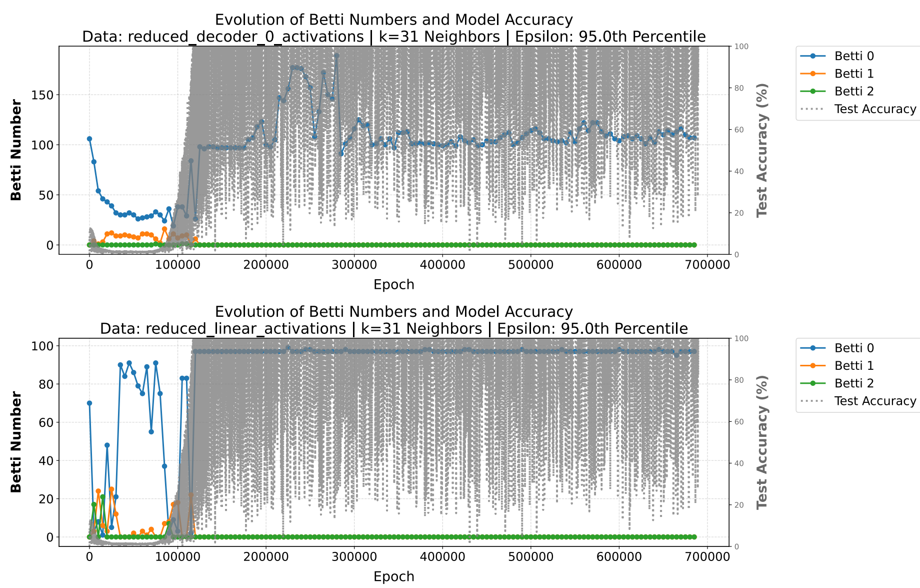

Fig. 8: **20% of training data.** In the linear layer, the same convergence as before is observed, although preceded by anomalous behavior. The decoder converged to a larger number of connected components (Betti 0) as generalization occurred.

Modular multiplication — partial generalizing conditions. Figures 9 reveal additional phenomena absent in the addition task. At $q = 15\%$, the model makes partial progress toward generalization and the linear layer oscillation amplitude decreases over time without fully converging, producing an intermediate topological regime consistent with an incomplete transition. Together these observations support **H1** and **H2** across both tasks.

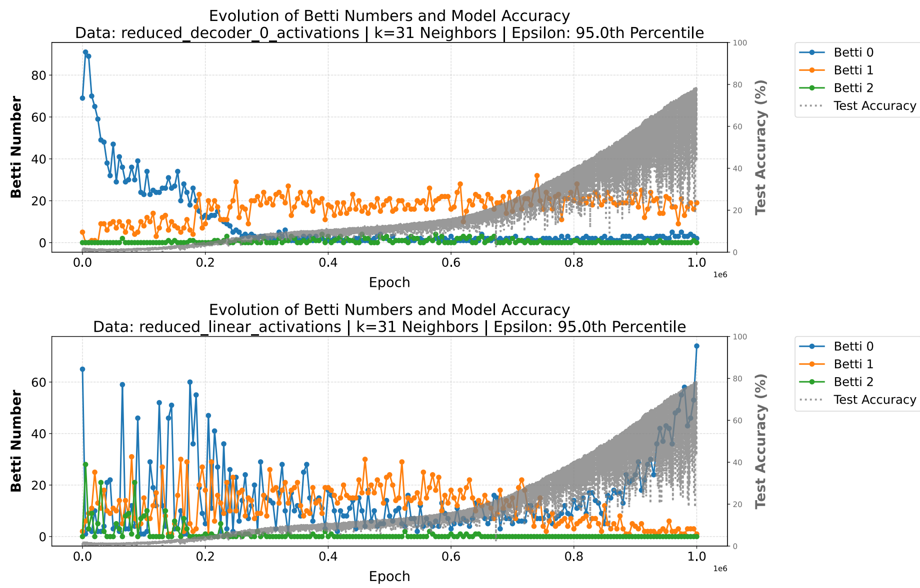

Fig. 9: **15% of training data.** The model exhibits partial generalization, accompanied by similar topological behaviors as noted in earlier successful runs.

4.2 Intrinsic Dimension Evolution

Modular addition. Figure 10 shows the evolution of $\hat{d}_t^{(\ell)}$ across all layers for all five training fractions. In generalizing conditions, intrinsic dimension decreases progressively across all layers, converging concurrently with the rise in validation accuracy. The compression is most pronounced in the first decoder block, where $\hat{d}$ falls from approximately 7–8 to values near 1–2 for $q \geq 20\%$. In non-generalizing conditions, intrinsic dimension decreases initially but stabilizes at higher values (near 6–7 for $q = 10\%$) and the gap between regimes widens over training, indicating that non-generalizing manifolds retain excess degrees of freedom not eliminated during training.

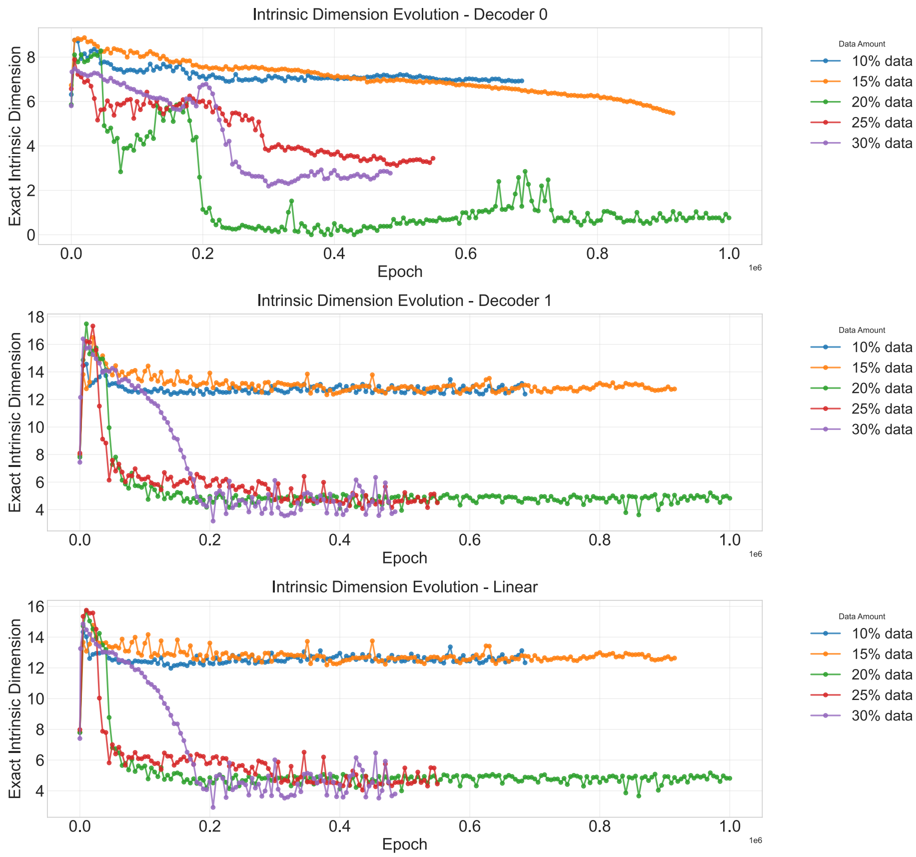

Fig. 10: Evolution of the intrinsic dimension of the representation manifolds for each model layer during training on the modular sum dataset.

Modular multiplication. Figure 11 shows analogous results for the multiplication task. The qualitative pattern is consistent with the addition task, with two differences: the separation between generalizing and non-generalizing conditions is more gradual, and non-generalizing conditions exhibit more sustained high-amplitude variation without clearly plateauing, consistent with the less stable Betti number dynamics reported in Section 4.1. Across both tasks and all ten conditions the evidence strongly supports **H3**: successful generalization is accompanied by intrinsic dimension reduction; failed generalization is not.

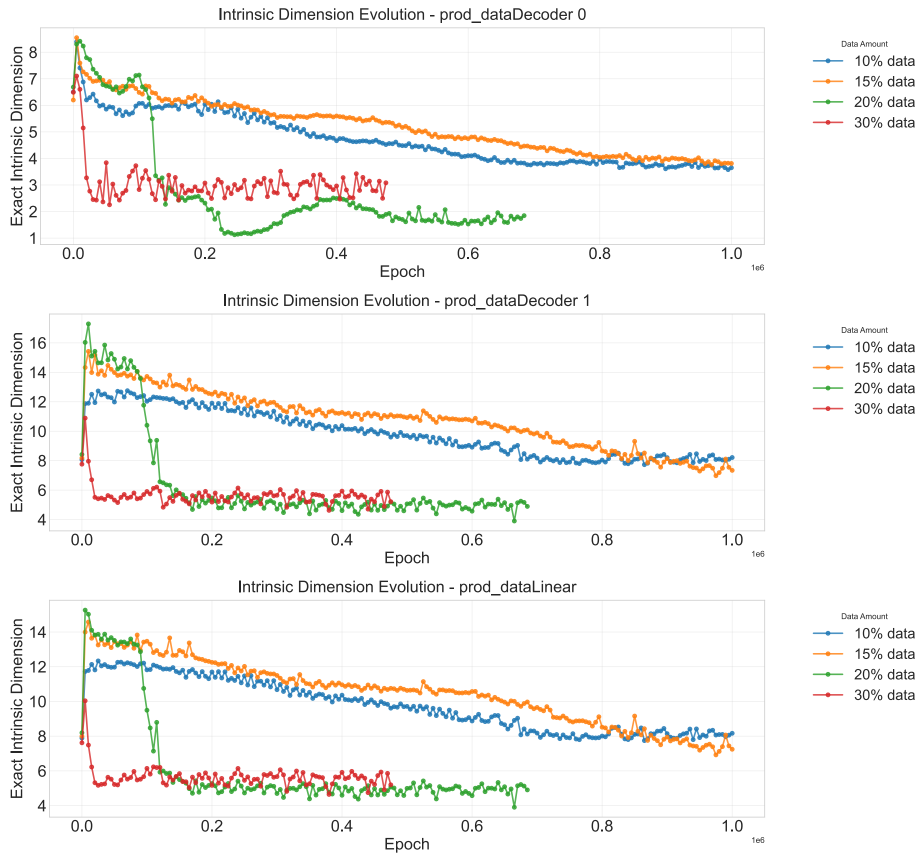

Fig. 11: Evolution of the intrinsic dimension of the representation manifolds for each model layer during training on the modular product dataset.

5 Conclusions

We applied Topological Data Analysis to characterize the grokking phenomenon in a small transformer trained on modular arithmetic tasks. Across two datasets, five training fractions, and four architectural layers, we found that successful generalization is strongly associated with a structural regularization of the learned representation manifold. Specifically, β_0 converges to a stable value in both the decoder and linear layers concurrently with the rise in validation accuracy; i.e. decreasing in the decoder as intermediate representations consolidate into fewer connected components, and increasing in the linear layer as output representations organize into well-separated clusters; while intrinsic dimension decreases progressively across all layers to values substantially lower than those observed in non-generalizing conditions. These patterns are robust across all ten experimental conditions, and the partial generalization observed in the multiplication task at $q = 15\%$ produces intermediate topological behavior, suggesting the transition is continuous rather than abrupt.

Several directions remain open. First, a systematic sensitivity analysis over the pipeline hyperparameters — the neighborhood size k, the distance percentile threshold, and the persistence noise cutoff τ — is needed to establish the robustness of the reported signatures under different construction choices. Second, the identified topological markers should be tested on algebraic extensions of the current setting, including other cyclic groups $\mathbb{Z}_n$, non-commutative structures such as permutation groups, and alternative modular operations. More broadly, since the TDA pipeline is task-agnostic, it could in principle be applied to other domains where delayed generalization has been observed, such as sparse parity learning or image classification; validating whether the same topological signatures accompany grokking beyond modular arithmetic remains a longer-term goal. Finally, the correlational nature of our findings motivates exploring topological regularization losses that directly encourage manifold simplification, which could serve both as a causal test and as a practical mechanism to accelerate or induce grokking.

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Disclosure of Interests. The authors declare no competing interests.

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